

Read quality in the PDAC KRAS dataset

WEEK 2 PROGRESS REPORT

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Interim version: 26 files reviewed; newly submitted jobs were still pending.

Wording revised on 23 September 2026. Results retain the original reporting stage.

Purpose

This coursework project compares control and KRAS small interfering RNA (siRNA) treatment in pancreatic ductal adenocarcinoma (PDAC) cell lines. This stage checks read quality and prepares paired files for genome alignment.

Work completed

The input list contains 34 compressed FASTQ files from 17 paired-end runs. At this stage, 26 FastQC 0.12.1 reports had been reviewed, covering 13 complete pairs. The available MultiQC 1.35 report contained 22 files.

Jobs submitted

On 22 September, eight remaining raw-read quality checks and a complete MultiQC report were submitted. Imports were also submitted for the four missing files from SRR24828471 and SRR24828472. HISAT2 2.2.3 pilots were submitted for Control M1 and KRAS P1 against human hg38, using unstranded, forward-stranded (FR) and reverse-stranded (RF) settings.

Each pilot used the first 1,000,000 read pairs in file order. Results from these submissions were not yet available at the cutoff for this interim version.

FastQC results for 26 files

FastQC check	Pass	Warn	Fail	Files
Per-base quality	26	0	0	26
Per-tile quality	26	0	0	26
Per-read quality	26	0	0	26
Base composition	0	0	26	26
GC distribution	26	0	0	26
N content	26	0	0	26
Read length	26	0	0	26
Duplication	0	0	26	26
Overrepresented sequences	26	0	0	26
Adapters	26	0	0	26

Each file contained 150-base reads, with 38,181,985-47,840,900 reads per file and 50%-52% GC. R1 and R2 counts matched within all 13 reviewed runs. The 22-file MultiQC report showed 57.0%-65.2% duplication.

Interpretation

All 26 reviewed files passed the base-quality and adapter checks. Base composition and sequence duplication were flagged in each file. Library preparation and differences in transcript abundance can affect these measures; the flags alone do not justify removing duplicate reads.

Work remaining at this stage

The submitted jobs still needed review. The next steps were to confirm the imported files, bring FastQC and MultiQC coverage to 34 files, assess strand specificity and run the full alignments. This interim version therefore contains no alignment percentages.

The pilot subsets were taken from the start of each file, rather than sampled at random. Separate run names do not establish independent biological replicates. No gene-count or differential-expression results were available.

References

1. FastQC reports reviewed on 19 September and Galaxy job submissions checked on 22 September 2026; MultiQC 108. Analysis records: github.com/yichuanlin07/pdac-kas-rnaseq.
2. Klomp JA et al. Defining the KRAS- and ERK-dependent transcriptome in KRAS-mutant cancers. Science. 2024. doi:10.1126/science.adk0775.
3. FastQC documentation. bioinformatics.babraham.ac.uk/projects/fastqc/.

Author contribution

Yichuan Lin organized the data and coordinated the analysis and report preparation.

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